

Measurement Paper

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1 Introduction

Online test-taking allows for a depth of click stream data regarding test behaviors. The value of this data form goes beyond test performance as patterns of behavior regarding test-taking behavior can also be deduced from response process data. Though response data generally has a very high precision level - specific test-taking actions are recorded with exact timing - this specificity can limit analysis. Ultimately, a level of coarseness must be introduced for understanding to be viable. Recent work in developing behavioral profiles includes creating sequences of item-level behavior profiles (i.e. one behavior per item, across a test) (Shen and Kanopa [5]). By creating item-level heterogeneous profiles and considering these across a test, an effective sequence of behavior can be formed.

By considering an item to be a fixed time period, we can think of these behavioral profile sequences as similar to Categorical Function Data (CFD) (Gabadinho et al. [1]). CFD considers sequences as stochastic processes with continuous time but finite states. In the case of our sequences of behavioral profiles, these states are the item-level profiles if we create some alphabet of states across the sequence. In other words, if we create item-level profiles but use one naming convention across the different items we can create sequences of behavioral profiles similar to CFD. In order to consider behavioral sequences in this framework, a strong assumption must be made that item lengths are comparable to time periods), such that at each time point (or new item) there is the potential for transition from one state to another.

Once these sequences are formed, groupings of test takers can be created using hard clustering. Life course analysis work, where previous behaviors are used to consider outcomes, heavily uses sequence dissimilarity distance metrics to cluster life courses. A large amount of life course analysis work is based on sequences of distinct non-overlapping states, with pre-defined time periods (for example, employment status for a given month of the year). This is largely the result of the need to introduce coarseness into life course analysis by coding life courses into fixed states at varying time periods (akin to CFD). There are inherent similarities between this and test latent behavioral sequences, as both areas have defined time-periods and, generally, a non-extreme number of states. As a result, there is a clear advantage to exploring the range of distance metrics used in life-course analysis within the context of latent behavioral sequences.

There are clear parallels between creating sequences of behaviors from item-level heterogeneous profiles and the work to make coarser representations in life course analysis. Distance metrics highlight the qualities of sequences differently; for example, some place greater emphasis on timing, duration, or sequencing. Bringing these metrics to behavioral sequence data could allow for a clearer understanding of different qualities in these sequences. Though there are parallels with life course data there are fundamental differences in structure that analyzing a range of clusterings could highlight.

Life Courses have a temporal factor that we expect to see less in this analysis - it is expected in life course work that people stay in the same state for much longer. There is also no inherent structure that all are following - though it could be argued that there is some structure in educational timing norms, this does not extend to the whole sequence as in a test sequence. There is also a fundamental difference in what development looks like in these types of sequences. In life course work we expect to see current states having a much heavier association with previously visited states. For example, we expect that time spent in education will have fundamental impacts on employment. There is also a fundamental ordering involved in life course work that we expect less in test behavior sequences. For example, we expect that employment comes after secondary education, not the inverse. Generally, the length of life course sequences carries with it an expectation of changed behavior that we do not expect within a test-taking time period. Students are unlikely to change their behavior massively over the course of one test. This makes clustering these test sequences more intuitive than with life courses, as we are not required to expect that clustering can capture great differences in behavior or timing of behavioral changes. Test-taking behaviors take on less dimensions than life courses.

In order to create a deeper understanding of our sequence representations it will also be interesting to look at the value of forgoing behavioral latent class analysis per-item and only consider differences in the underlying tool use profiles. A potential extension of this would be to consider a distance metric that considers both the encoded latent behavioral states as well as the distance between the underlying tool profiles at every item.

Though there are clear links between created behavioral sequences and life course sequences, the fundamental structural differences can be exposed by looking at the clusterings formed by differing distance metrics. This comes as a result of the variation in emphasis found across the distance metrics used in Sequence Analysis generally. For example, some metrics place a greater emphasis on timing of events while others place more value on similarity of spell durations (i.e. how long the state remains unchanged). As behavioral sequences based on item-level profiles are strictly limited to the form of the test and it's pre-determined item ordering, it is intuitive to assume variation in the form of the clusters based on distance metric. In particular we expect to see metrics that emphasize spell duration to be less effective.

1.1 National Assessment of Educational Progress Data

The Nation's Report Card (otherwise known as the National Assessment of Educational Progress) is a nationally administered test of student academic ability. Provided by Educational Testing Service for the NAEP 2019 Data Mining Competition (Patikorn et al. [4]), the data are sample of student's response processes from the 8th grade math assessment. Extensive work has been done analyzing this data, primarily with the focus of evaluating the efficiency of test takers. Test data is split into two blocks - the first generally to be used in training for prediction of performance in the second block. However, in this work we are less interested in predicting efficacy of test taking but rather in finding groupings test taking behaviors. To create these groupings we will primarily focus on tool use throughout the test. The online environment for administering NAEP includes tools for math calculations that students are freely able to use (National Center for Education Statistics [3]).

The following tools can be used:

- Color Contrast/Change Theme
- Zooming
- Text-to-Speech
- Scratchwork

- Equation Editor
- Calculator

For specificity of analysis we will consider the latter four areas, as the adding in the complexity of the visual tools may add a dimension to this work that is too detailed for our comparison of behaviors. Keeping to only tools that aide in doing the actual math work makes the domain specific enough that we hope to find well formed groups.

As click-stream data, the exact moments of behaviors are recoded at a very fine level, with every action in the test recorded precisely. There are general actions, such as click "Next Item" as well corresponding more specific details about the specific user's action.

The following variables are included:

- StudentID - A unique identifier of student
- Block - The block of the test (we have access to "A")
- AccessionNumber - A unique identifier for the test item
- ItemType - Question types (e.g. multiple choice)
- Observable - Specific action taken (e.g. "Open Calculator")
- ExtendedInfo - Details into the Observable
- EventTime - The exact time of the action.

This is not an extensive list but the data takes on a "long" form, meaning that there depth captured within these variables - particularly in the "ExtendedInfo". Our aim is to consider item-level heterogeneous profiles, so we form tool use profiles for each item. Tool uses are defined in the "Observable" variable, and we maintain this coarseness for ease of analysis. We will take only the training block and consider student behavior across the 19 items in the 30 minute time limit. There are 1232 unique students in the dataset.

For behaviors to be comparable across students we only consider students who saw every item on the test. All students received the items in the same order but were able to return to items. This has the potential to introduce a high amount of complexity to the process as item ordering becomes highly individual. To create comparable sequences we make the strong assumption that there is an inherent value in the order in which the items were first seen. This is the structure of the test. However, we do not limit tool use to behavior on the first viewing of an item. Students may come back to an item and use a tool they did not in the first pass. This does create inherent complications about what these sequences actually represent - but, again, creating some coarseness is necessary here for comparisons of behavior.

After limiting to students who saw every item in the correct order, we have 952 unique students. There are then 42 unique "Observables", including tool uses as well as more general actions such as item transition movements. To focus on the tool use behavior of student test takers we limit the Observables to the following:

- Draw - Use the drawing tool provided in the Scratchwork editor
- Erase - Use the erase tool provided in the Scratchwork editor
- Open Calculator
- Eliminate Choice - Eliminate one of the available answers (if possible)
- Highlight - Use the highlight tool provided in the Scratchwork editor

- TextToSpeech - Use the Text-to-Speech function
- Equation Editor Button - Use the onscreen equation editor to enter numbers and expressions using the available mathematical symbols.

There are more tool uses in the list of observables (e.g. "Scratchwork Mode On/Off" or "Close Calculator"). We choose to limit the tools to only those that are direct use actions, rather than about getting into a state where the tool can be used or put away. We find that direct action Observables are more relevant and it is unlikely that latent class analysis could capture nuances with the limited data.

AccessionNumbers map to unique test questions and can take on the following ItemTypes, which indicate the form of the question:

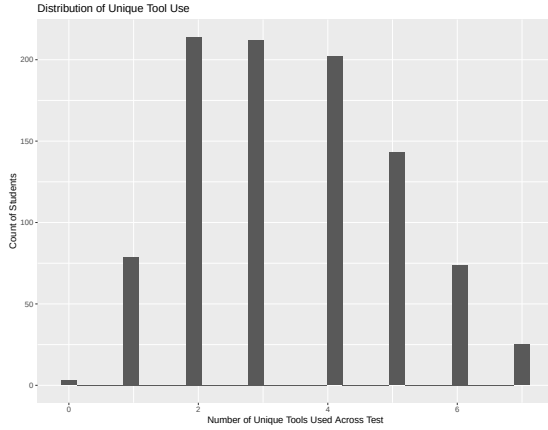
- MCSS - Multiple-choice single-selection question.
- MatchMS - Match multiple-selection question. Students drag different values to multiple pre-defined blanks/locations to complete expressions, equations, etc.
- FillInBlank - Single fill in the blank question.
- MultipleFillInBlank - Multiple fill in the blank question.
- CompositeCR - Composite constructed response question. Students fill in one or more blanks, usually open ended. This could be paired with other minor questions e.g. drop down

There are also AccessionNumbers for the directions page and help page, but as these are not test questions they are removed. All tools are available for all items with the exception of the Eliminate Choice, which was not available in items 4, 7, 9, 12 or 13 - all other questions are MCSS. Also, the "Equation Editor" button was only used by students in question 1, 3, 6, 7, 9, 11, 12 and 13.

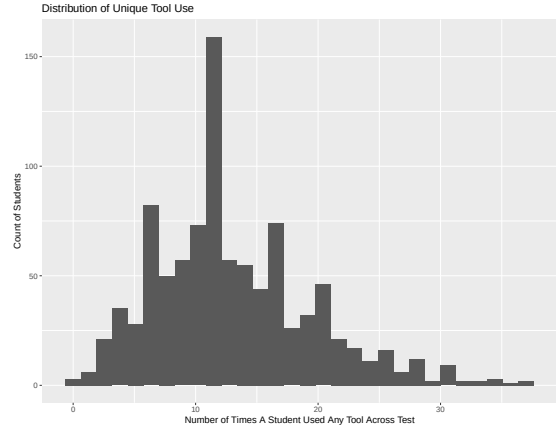
As mentioned, though we enforce the structure of test (i.e. item ordering) onto the sequences, tool profiles are created as binary variables for each relevant tool's use at any time across the test taking period. It is possible to create counts for tool use on specific items but the coarseness of general use is more appropriate for this analysis. It also isolates types of behavior without having to include the complexities of individual repeated click behavior (i.e. going in and out of drawing mode).

We therefore end with 952 students, 19 Accession Numbers and 7 tool uses. It should be noted that because the data takes on a click stream form, in creating a full set of binary tool use profiles for every students a large amount of filling with 0s (phrasing) was required. Of the 133280 unique student-item-tool combinations, there are only 12471 cases of a tool being used.

The figures below show the distributions of student tool use. Figure 1a. shows the distribution of number of unique tools used by students while Figure 1b. shows the distribution of number of times a particular student used any tool.



(a) Figure 1a.



(b) Figure 1b.

Though the tool use data is very sparse, there is clear evidence that the tools are used systematically across the test by students. However, with the knowledge that questions take on different forms (i.e. are of different ItemType) it is reasonable to consider the difference in tool use by item. Figure 2 below shows the proportion of students who used each tool for each item in the test.

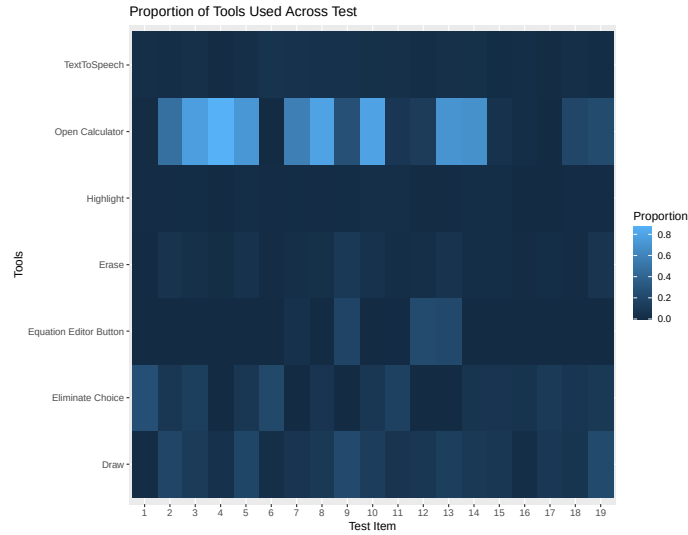


Figure 2: Figure 2

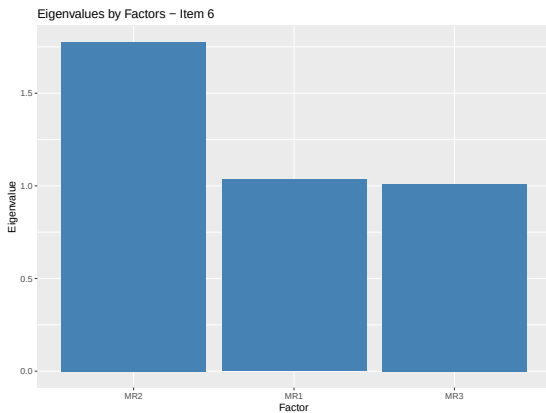
From the ItemTypes and Figure 2 we can see that there are structural difference in tool use by item. This will inherently alter behavior for these items and must be considered when we are creating our alphabet of behavioral states. However, we can see that, on average, tools are not used very frequently. This may be a potential issue for our analysis, given that variance in tool use will be a driver of distinguishing behavioral patterns.

2 Dimensionality Reduction

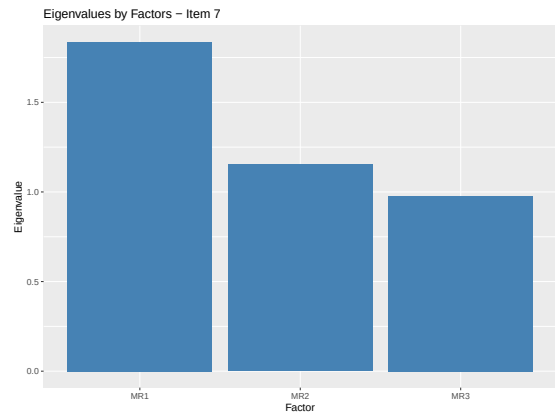
The analysis of Figure 2 makes it clear that there are indeed structural differences in tool use across items based on item form. An argument could be made that instead of creating item-level heterogeneous profiles the full set of tool profiles (i.e. every student-item combination) could be evaluated together in one dimensionality reduction. This would inherently eliminate the idea of nuance between items, which we know to exist. It would also remove any idea of timing in tool use, as all tool use choices would be assumed to be considered simultaneously.

Though the aim of this paper is to consider the latent behavioral profiles on an item-level, we do still need to evaluate whether it is appropriate to compare behaviors across items. Looking at a per-item factor analysis on two tool profiles can give us some insight into whether this is appropriate. For example, if items have very different optimal number of factors it may not be easy to qualitatively match the behaviors across the items. Performing a factor analysis let's us check how many factors are reasonable and this may map to the number of latent classes. We chose factor analysis over a principal components analysis because it is not unrealistic to assume that there would be correlations among tool use behaviors.

Using Figure 2 as evidence, we will look at the difference in factorizing items 6 and 7. These are far enough into the test that it can reasonably be assumed that students have considered tool use and comes after a block of heavy calculator use. These two are chosen specifically because item 6 has low calculator use and item 7 has high calculator use. Below are the scree plots for these two items - both highlighting that really there is one dimension which captures a significant amount of variance in tool use. This is a concern for our analysis, as it may indicate that we will not be able to find nuances in tool behavior. We chosen to retain 3 factors in order to, hopefully, capture some nuance in the difference of these two factorizations (even if they eigenvalues do not indicate significantly amount of variance explained).

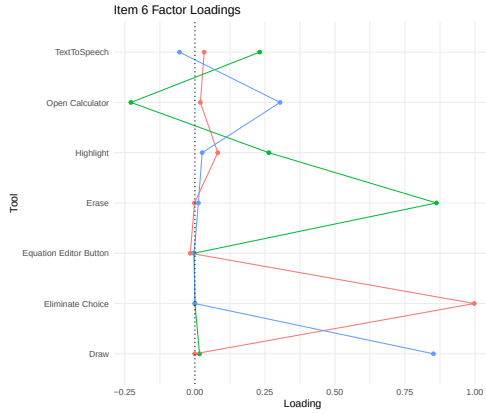


(a) Figure 3a.

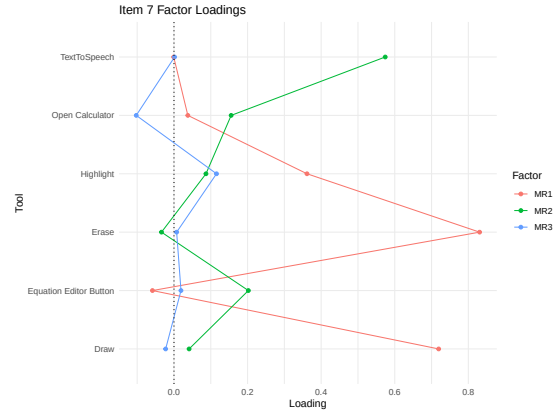


(b) Figure 3b.

The factor loadings shown in Figures 4a and 4b show that the tool use profiles associated with items 6 and 7 vary differently. The highest eigenvalue factor for item 6 (MR2) loads heavily onto the "Eliminate Choice". On the other hand, the highest eigenvalue factor for Item 7 loads onto "Erase", "Draw" and "Highlight" with the greatest magnitude. These two factorizations capture different tool use patterns and it would not be reasonable to consider the unlying items as similar enough to ignore the structural differences.



(a) Figure 4a.



(b) Figure 4b.

Something to note is the fact that calculator use does not load very strongly on any factor in either of the factorizations. This is an inherent function of doing a factor analysis because we know that there is a difference in baseline calculator use between these two items but that between item variation cannot be captured by an item-level factor analysis. The variance focus of factor analysis means that instead of coming baseline level of behavior we capture differences in behavior departing from that baseline. Instead, turning to an item-level Latent Class Analysis allows for a more intuitive connection between behavioral states.

As a note, though we know that it is not reasonable to assume item similarity, we cannot deduce from this that there is any implicit value in item ordering. Instead looking at distance metrics after creating behavioral sequence could give us more insight.

3 Latent Class Analysis For Behavioral Profiles

In order to connect item-level heterogeneous profiles it is important to maintain an understanding of item-level "norms". Performing latent class analyses at an item-level allows us to consider the general behavior groupings of students at specific items while providing nuances in behavior. We can then define an alphabet of behavioral states that will be assigned to the item level classes but are common between items. Effectively we map latent classes to overall behavioral states across the test, creating sequences of behavioral states where the states are common among items.

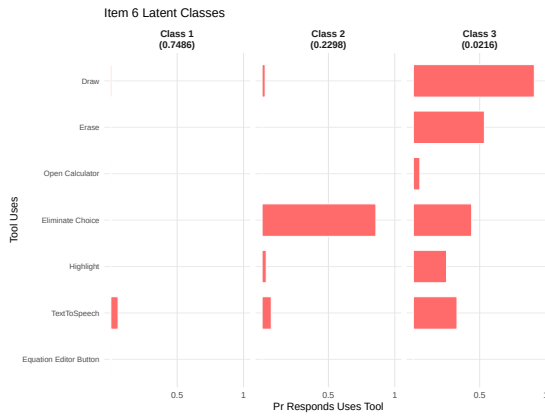
The factor analysis above highlights that there are definitely discrepancies in behavior across items, so we should be cautious in our assignment of variables. To appropriately represent behaviors at the item level there is a necessary balance between creating a very large alphabet of states that capture item differences and a very small alphabet that is too coarse to capture item nuance at all. A similar balance is required in selecting the number of latent classes per-item. Ultimately, the alphabet is the result of being generous with the number of latent classes per item (generally opting for a higher number of classes where possible) but fairly terse with the number of unique states in the alphabet. In general we find an average of 3 latent classes per item and map these classes to 7 states in our alphabet, encoded as follows:

- N - None
- EC - Eliminate choice focused
- D - Drawing focused

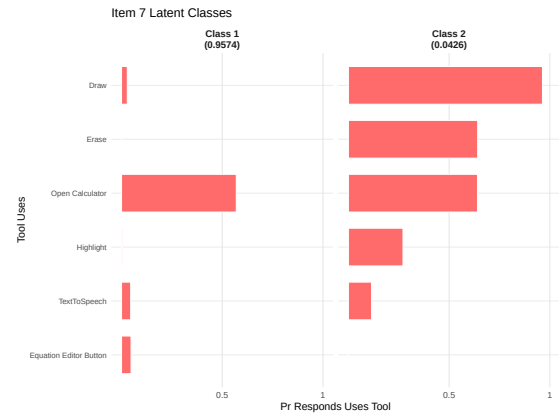
- TS - Text to Speech focused
- C - Calculator use focused
- EE - Equation Editor focused
- H - Highlighter use focused

Though these map to specific tools, the "D" drawing state contains general use of "Draw" and "Erase", with a lower level of "Highlight". However, the "H" state is included because there are items where highlighting maps to groups distinctly from drawing. Generally, the number of latent classes selected was determined by minimizing BIC and AIC while maximizing log-likelihood and entropy. However, this evaluation included evaluating the form of the plots themselves and this generally lead to picking a higher number of classes and forgoing some fit statistic optimization. Ultimately this choice is justified because generally only two classes would be chosen - the baseline (Calculator or not) and a general other tool use. This is not that interesting inherently so additional classes were added in hopes that this could induce more complex behavior profiles with greater nuance.

Full plots for the latent class analyses can be found in the appendix, but the two plots below are useful to explain the thought process behind coding items that have differing baseline calculator use. Again, we will look at items 6 and 7, because of the distinction in Calculator use for those items.



(a) Figure 5a



(b) Figure 5b

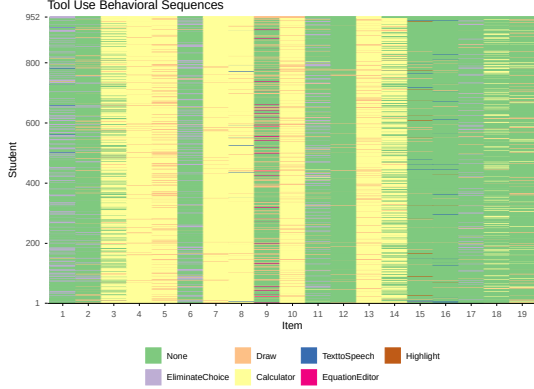
Figure 5: Latent Classes for Items 6 and 7

Item 6 had very low, proportional, Calculator usage whereas Item 7 had a fairly high usage. The three classes found in Item 6 are coded as "N", "EC" and "D", whereas the three classes found in Item 7 are coded as "C" and "D". Being able to capture this distinction in levels of Calculator usage and tool use-difference by item is an advantage of the latent class approach. Qualitatively coding (as we do here) requires these underlying structures to be accessible.

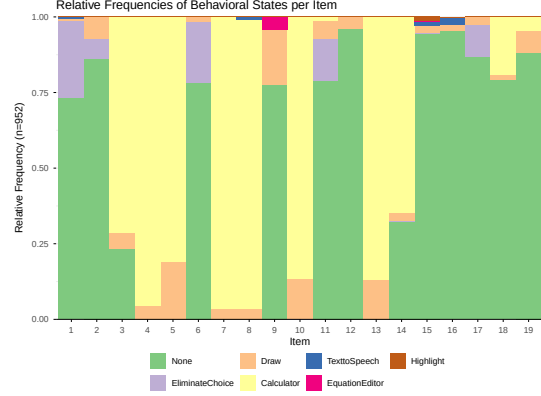
4 Sequences

Once these behavioral profiles are created we can construct sequences wherein the behavioral state at each item corresponds to it's index in the resulting sequence. As discussed, this reduces the complexity of the response processes but allows us to consider the impacts of different distance

metrics. There is also an argument to be made in favor of coarse representations of sequences - the complexities and fineness of sequence data is subject to be effective noise (Lundberg et al. [2]).



(a) Figure 6a



(b) Figure 6b

Figure 6a displays all 952 student sequences. There is a clear parallel between the calculator use prevalence shown in Figure 2 and the higher frequency of membership in the "C" (Calculator use) state in Items 3, 4, 5, 7, 8, 10, 13 and 14. We see this both in that states dominate the full set of sequences and the prevalence of those states proportionally (as shown in Figure 6b).

Though this basic distinction is caught by our qualitative coding, looking at the general distribution of the behavioral states across the items is not encouraging if we want to find real nuance in tool use beyond this calculator use baseline and other general tool use. Regardless, we will look at how these sequences cluster with different distance metrics to evaluate our intuitions about the qualities of the behavioral sequences as compared to typical life course sequences.

5 Clustering with Sequence-Based Distance Metrics

5.1 Sequence-Based Distance Metrics

Ultimately, our aim is to create clusters of test-taking behavior types. A main source of evaluation of distance metrics used in Sequence Analysis is "What Matters in Differences Between Life Trajectories: A Comparative Review of Sequence Dissimilarity Measures" (Studer and Ritschard [6]). The paper argues that there are three main types of distance metrics for sequences:

- Distances between distributions
- Measures based on counts of common attributes
- Edit distances

Distances between distributions consider the difference in time spent in each state within the distribution. The dissimilarity is measured by the difference between state distribution vectors. Here we consider both Euclidean distance and Chi-Squared distance between these state distributions. As the focus is time spent in each state (regardless of order), these metrics are insensitive to order and timing. Distances between distributions are mainly focused on trends of states generally.

Measures based on common attributes may be the most appropriate for the behavioral sequences because of their focus on specific period-wise similarities. Hamming distance counts the number of

positions with non-matching states. It can only be used with sequences of the same length because it is a position-wise comparison. But the nature of test sequences means that Hamming distance’s sensitivity to timing mismatches (in life course entering a job a month apart, say) is not a problem but rather a benefit. Hamming seems like the most intuitive distance metric for our sequences. A second common attributes measure we consider is length of Longest Common Subsequence (LCS). This is effectively the number of elements that can be uniquely matched. For life course work this allows the flexibility of the same subsequences but with differing start points. Mismatched timing is not a concern for test sequences but the structure of the test could enforce specific subsequences that reflect similarity in user behaviors. For example, if two students had the exact same calculator and non-calculator usage from item 3 to 14 we would assume them to be very similar test takers.

Finally, there are measures based on edit distances. Generally these fall under the category of Optimal Matching which can be described as the minimal cost of transforming one sequence to match another with some sort of cost. Transformations include substitutions, insertions and deletions and Optimal Matching generally can be seen as a weighted sum of time shifts (insertions or deletions) and a weighted sum of mismatches (substitutions). If no insertions or deletions are allowed, Optimal Matching effectively becomes Hamming distance as it is based on substitutions alone (particularly if the sequences have the same length, enforcing pairwise comparison). Given that we know that time shifts are not relevant to test behavior sequences it is highly unlikely that an Optimal Matching distance with an insertion/deletion focused cost will be the most effective in this data. Generally, high insertion/deletion costs makes the metric more sensitive to timing. We consider Optimal Matching with Transition Rates (OM-Trate) cost where substitution costs are derived from the frequency of transition between states across the data. This places harsher emphasis on rare states, which could be a problem for our data because we are trying to find nuances within the rarer tool use states. As we know insertions and deletions are not relevant to our data, looking at Optimal Matching with an insertion-deletion focus is not useful.

As a summary we consider the following distance metrics:

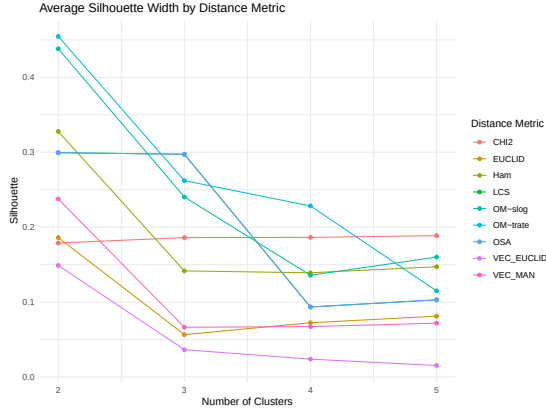
- Euclidean (Distance between distributions)
- Hamming
- Optimal Matching with Transition Rate Substitution Costs

5.2 Tool Profile Only Distances

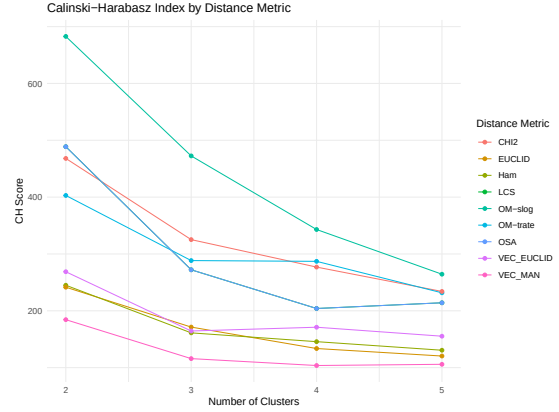
As a sanity check on the value of creating latent behavioral sequences from this tool profile data, we will also consider two distance metrics that are completely ignorant of the qualitatively coded sequences. We will look at both the simple euclidean and manhattan distances between tool profiles across all items. We treat the tool profiles as completely ignorant of item order and simply find the distance between all item tool profiles in the form of a single vector.

5.3 Comparing Distance Metric Performance

To compare distance metrics we use Agglomerative Hierarchical Clustering (AGNES) with a ward linkage for all distance metrics. This is frequently used in life course analysis and therefore seems appropriate for sequence-focused clustering methods.



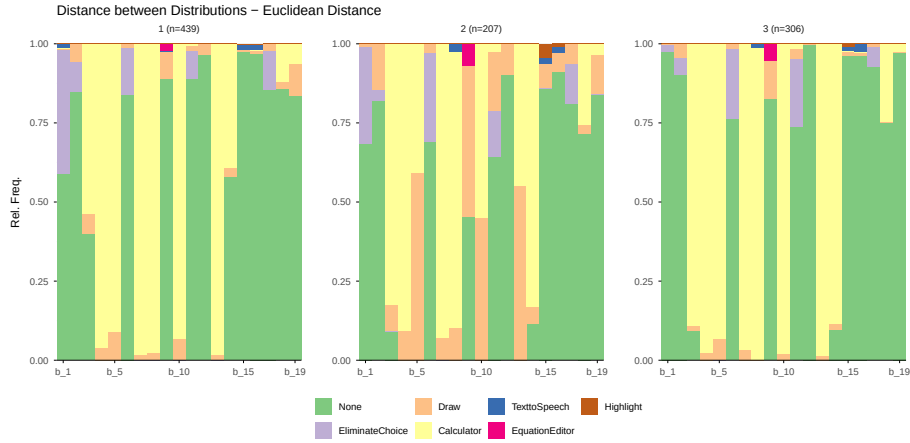
(a) Figure 7a

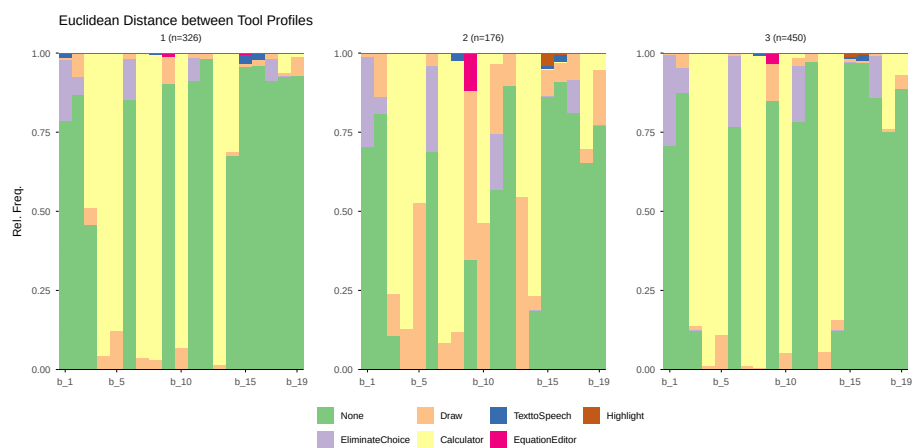
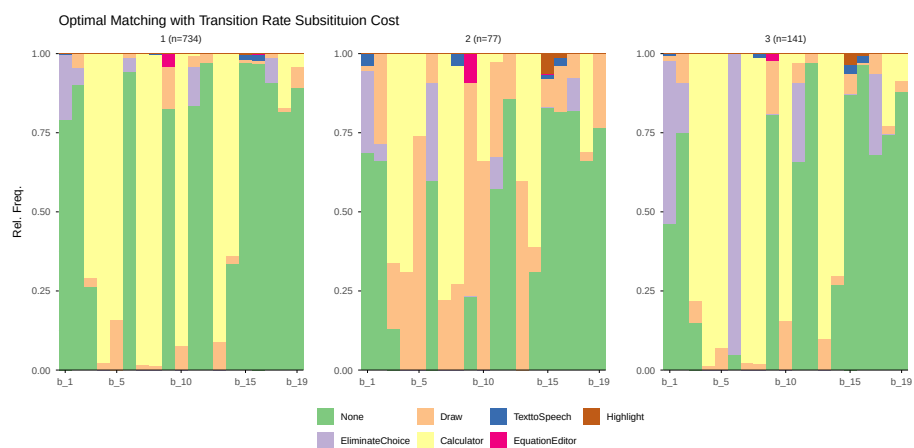
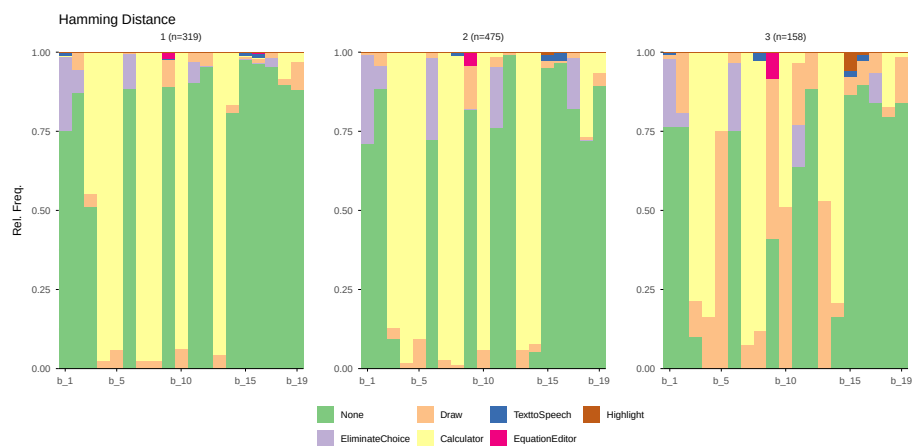


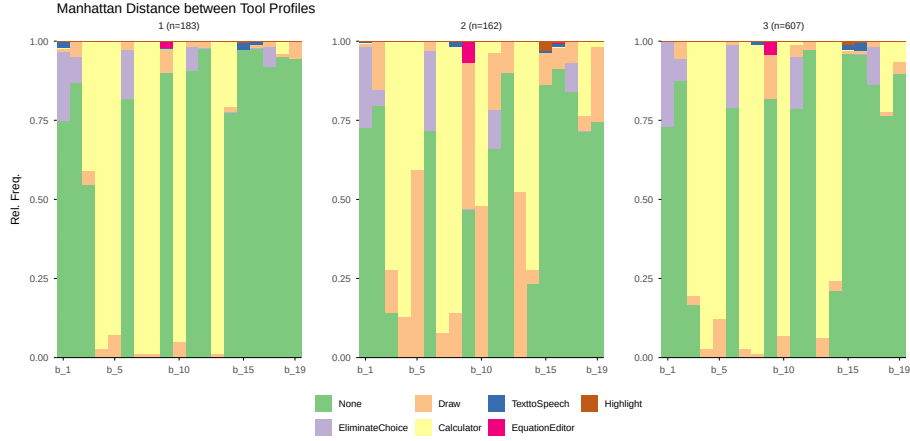
(b) Figure 7b

For every distance metric used in clustering both the average silhouette width and the Calinski-Harabasz index point to a two cluster solution. From the distribution of the data, and as discussed, this is likely because there is not enough nuance in the difference of non-general tool use. Finding nuance in tool use behavior with two clusters is not possible, so we look at the three cluster solutions for the selected distance metrics to evaluate more potential difference.

The figures below show the three clustering solutions for the different distance metrics. Though the sequence ignorant methods (euclidean and manhattan distance on full tool profile vectors) are not concerned with the behavioral sequences, we use the sequences that correspond to those underlying tool profiles to visualize the difference in clustering. This is method at least allows us to make comparisons, though does ignore some major features of the sequences.







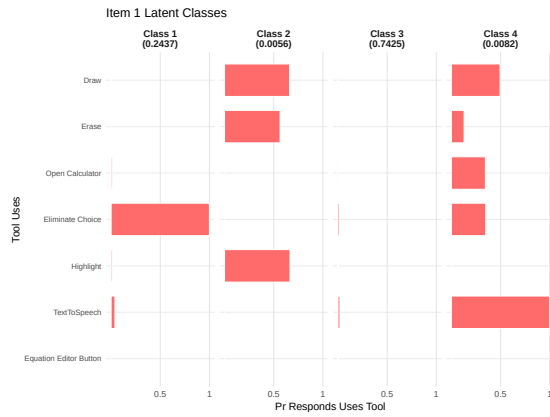
6 Conclusion

Because we are qualitatively coding the sequences it is important to maintain actual behaviors (rather than variance in behavior), so we have to use a method that maintains actual behavior rather. Qualitatively coding necessitates a baseline understanding of behavior in a certain item. Quantitatively coding may not be as limited as this!

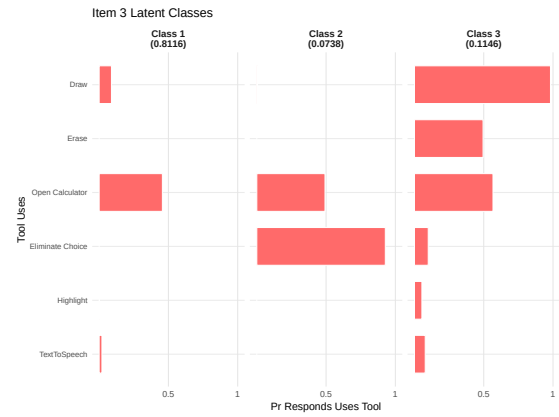
References

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7 Appendix

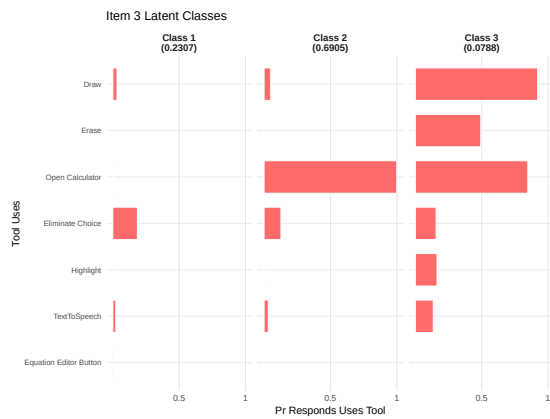


(a) Figure A1

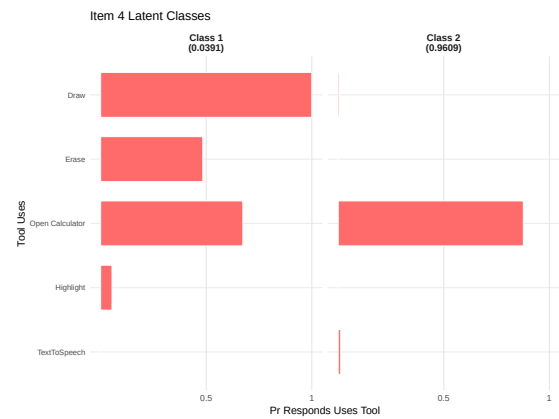


(b) Figure A2

Figure 8: Latent Classes for Items 1 and 2

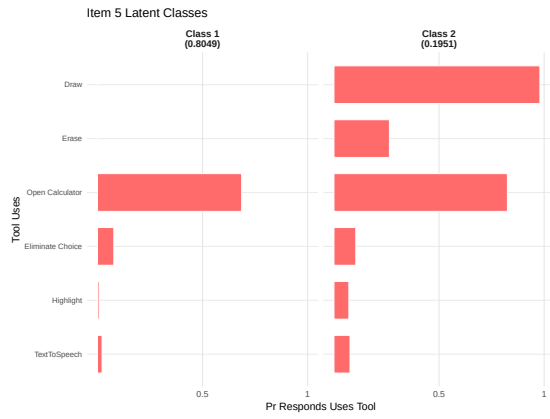


(a) Figure A3

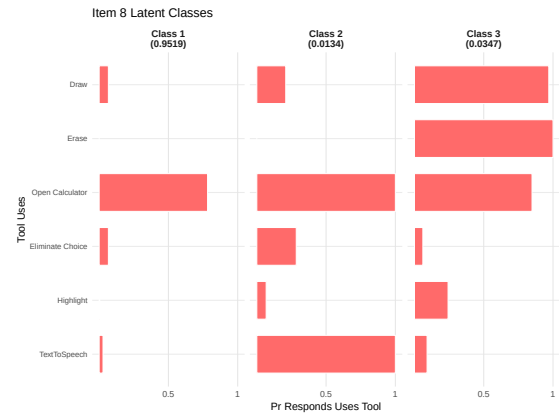


(b) Figure A4

Figure 9: Latent Classes for Items 3 and 4

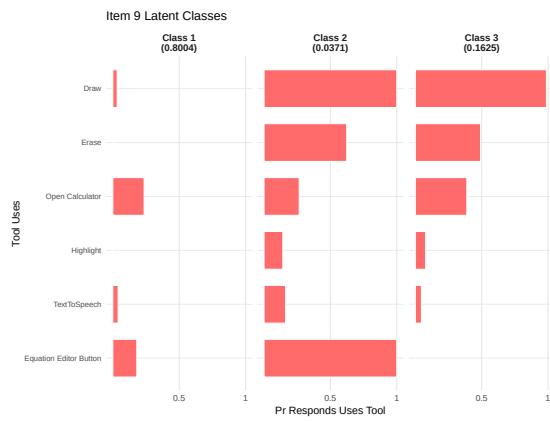


(a) Figure A5

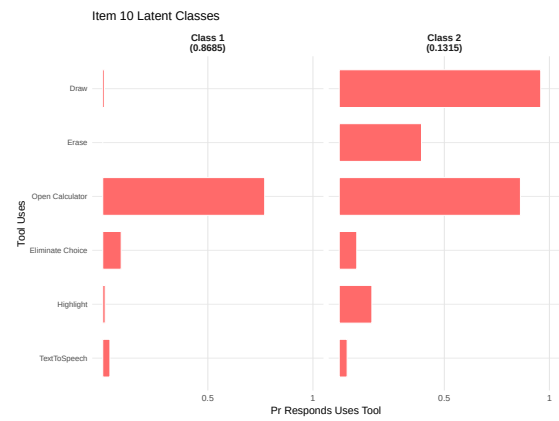


(b) Figure A6

Figure 10: Latent Classes for Items 5 and 8

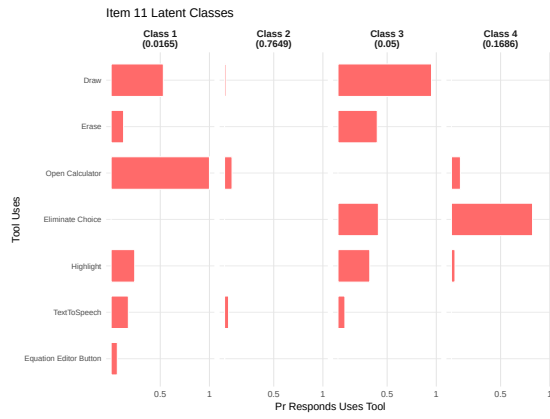


(a) Figure A7

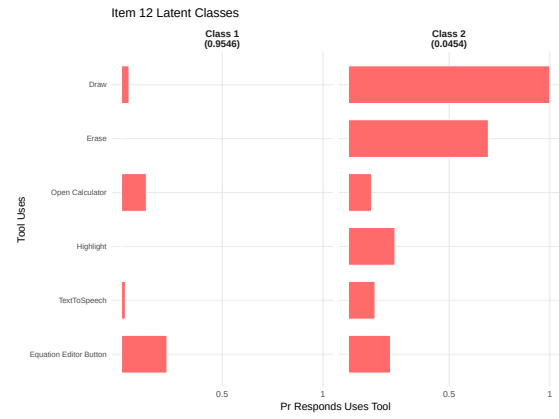


(b) Figure A8

Figure 11: Latent Classes For Items 9 and 10

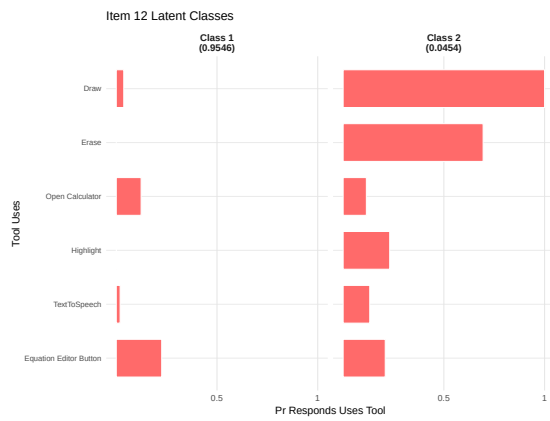


(a) Figure A9

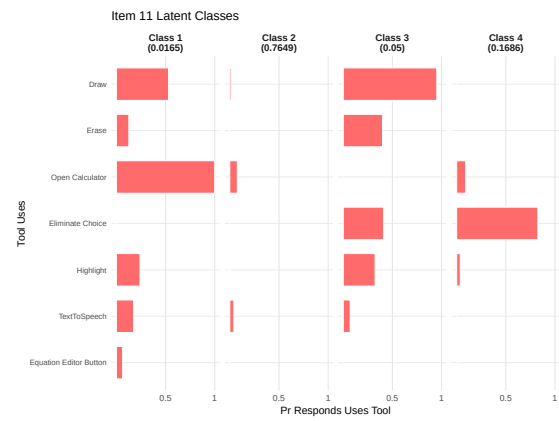


(b) Figure A10

Figure 12: Latent Classes for Items 11 and 12

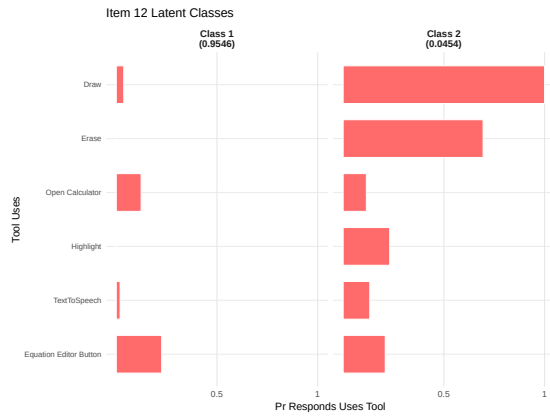


(a) Figure A11

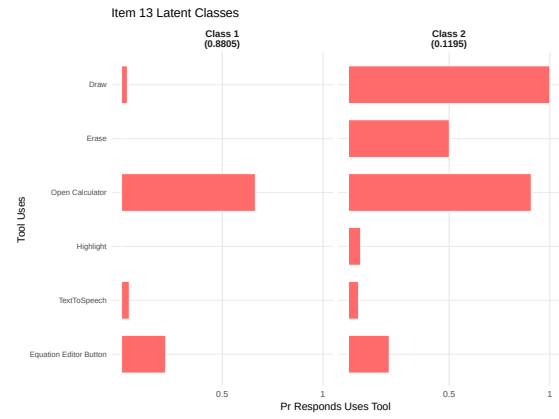


(b) Figure A12

Figure 13: Description for plots 13 and 14

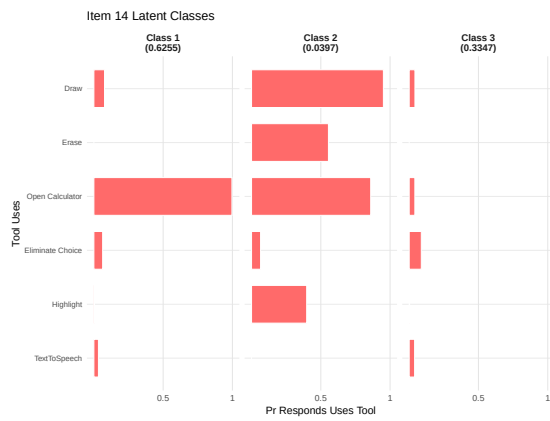


(a) Figure A13

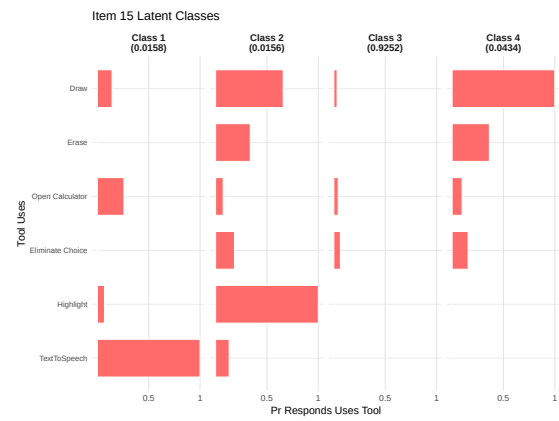


(b) Figure A14

Figure 14: Description for plots 15 and 16

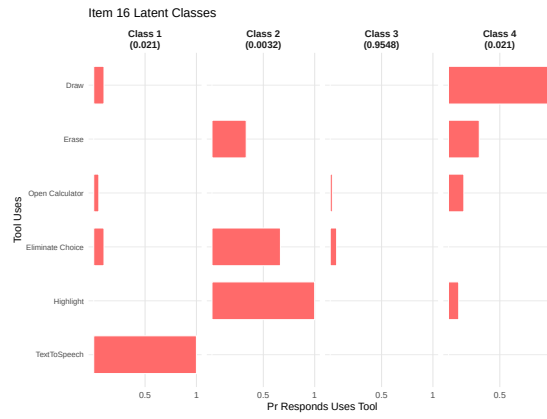


(a) Figure A15



(b) Figure A16

Figure 15: Description for plots 17 and 18



(a) Figure A17

Figure 16: Latent Classes for Item 19